

Changes in the microclimate around hydroelectric power station reservoirs and their association with worsening prognosis in patients with chronic heart failure

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Abstract. The construction and operation of large hydroelectric power station (HPS) reservoirs significantly alter local environmental conditions, leading to persistent changes in the microclimate of adjacent territories. These modifications include increased air humidity, reduced diurnal and seasonal temperature variability, more frequent fog formation, and stagnation of air masses due to diminished wind velocity. Emerging evidence suggests that such microclimatic shifts may adversely affect vulnerable populations, particularly patients with chronic heart failure (CHF). This study investigates the association between microclimatic changes induced by HPS reservoirs and clinical outcomes in CHF patients residing in nearby regions. A retrospective cohort analysis of 1,248 CHF patients (NYHA class II–IV) living within 15 km of three major HPS reservoirs in Russia was conducted over a 36-month period (2022–2025), with a matched control group from non-reservoir regions.

1 Introduction

Hydroelectric power stations (HPS) are a cornerstone of renewable energy strategies worldwide, valued for their capacity to generate electricity with low greenhouse gas emissions [1]. However, the creation of large reservoirs associated with HPS projects entails profound and often irreversible alterations to local ecosystems, hydrology, and—critically—the microclimate of surrounding regions. These anthropogenic changes include elevated relative humidity, reduced wind velocity, thermal buffering (leading to milder winters and cooler summers), increased fog frequency, and greater atmospheric stability. While the geomorphological and ecological consequences of reservoir construction have been

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extensively documented, the potential implications of these microclimatic shifts for human health—particularly among individuals with pre-existing chronic conditions—remain underexplored [2].

Chronic heart failure (CHF) is a complex clinical syndrome characterized by impaired cardiac function, reduced exercise tolerance, and heightened sensitivity to environmental stressors [3]. It is well established that meteorological factors such as temperature extremes, rapid barometric pressure fluctuations, and high humidity can trigger hemodynamic instability, fluid retention, and autonomic dysregulation in CHF patients, leading to disease exacerbation and hospitalization. Recent studies in environmental cardiology have begun to link specific climatic conditions to adverse cardiovascular outcomes; however, research has largely focused on macro-scale weather patterns or urban heat islands, with little attention paid to localized, human-made microclimatic modifications [4].

In regions surrounding large HPS reservoirs, the convergence of high humidity, stagnant air, and frequent low-pressure systems may create a uniquely unfavorable environment for cardiovascular stability [5]. Preliminary clinical observations and regional health statistics from reservoir-adjacent territories in Russia, Canada, and Brazil have reported elevated rates of cardiovascular morbidity, prompting concerns about a potential environmental–clinical nexus. Yet, to date, no comprehensive epidemiological study has systematically examined the association between reservoir-induced microclimatic changes and the prognosis of chronic heart failure.

This study addresses this critical knowledge gap by investigating whether residence in the microclimatic zone of large hydroelectric reservoirs is associated with a higher risk of disease decompensation, hospitalization, and arrhythmic events in patients with chronic heart failure. By integrating meteorological monitoring data with longitudinal clinical records, we aim to identify specific microclimatic parameters that may serve as independent predictors of poor outcomes, thereby informing both public health policy and environmental impact assessments in future hydropower development.

2 Methodology

This study employed a retrospective cohort design with concurrent environmental monitoring to investigate the association between reservoir-induced microclimatic conditions and clinical outcomes in patients with chronic heart failure (CHF). The research was conducted from January 2022 to December 2024 in three regions of the Russian Federation hosting large hydroelectric power stations (HPS): the Volga River basin (Volgograd Reservoir), the Angara River basin (Bratsk Reservoir), and the Kama River basin (Kama Reservoir). Each reservoir exceeds 1,000 km² in surface area and has been operational for over 40 years, ensuring stable microclimatic conditions [5].

Study Population. The exposed cohort comprised 1,248 adult patients (aged ≥ 40 years) diagnosed with stable CHF (New York Heart Association functional class II–IV) who had resided within 15 km of a reservoir shoreline for at least two years [6]. The control group consisted of 1,248 matched individuals with CHF living in comparable urban or rural settlements located >100 km from any major water body (>100 km²), selected using propensity score matching based on age (± 3 years), sex, CHF etiology (ischemic vs. non-ischemic), comorbidity burden (Charlson Comorbidity Index), socioeconomic status (education, income quartile), and baseline medication regimen (ACE inhibitors/ARBs, beta-blockers, diuretics, MRAs) [7].

Data Collection. Clinical data were extracted from electronic medical records of regional cardiology centers and included [8]:

- Frequency and causes of hospitalizations;
- Episodes of acute decompensation (defined by weight gain ≥ 2 kg in 3 days, worsening dyspnea, or need for intravenous diuretics);
- Documented arrhythmias (atrial fibrillation, ventricular tachycardia);
- All-cause and cardiovascular mortality.

Environmental data were obtained from 12 automated meteorological stations (6 near reservoirs, 6 in control zones), recording hourly measurements of:

- Ambient temperature ($^{\circ}\text{C}$);
- Relative humidity (%);
- Atmospheric (barometric) pressure (mmHg);
- Wind speed (m/s);
- Fog occurrence (binary).

Data were aggregated into daily averages and linked to individual patient outcomes via residential postal codes.

Statistical Analysis. Descriptive statistics summarized baseline characteristics. The primary outcome was time to first CHF exacerbation [9]. Cox proportional hazards models were used to estimate hazard ratios (HR) for adverse events in the exposed vs. control groups, adjusted for confounders. Additionally, conditional logistic regression assessed the short-term risk of decompensation associated with specific microclimatic thresholds (e.g., humidity $\geq 70\%$, pressure ≤ 740 mmHg) within a 72-hour window prior to the event. Sensitivity analyses excluded patients with renal failure or uncontrolled hypertension. All tests were two-sided, with statistical significance set at $p < 0.05$. Analyses were performed using R v4.3.2 and SPSS v28.

Ethical Considerations. The study was approved by the Institutional Review Boards of participating institutions (Protocol No. HPS-CHF-2021-07). Patient data were anonymized; informed consent was waived due to the retrospective design.

3 Results

1. Baseline Characteristics of the Cohorts

Following propensity score matching, the exposed and control groups were well balanced across all measured demographic, clinical, and socioeconomic variables (Table 1). The mean age was 63.4 ± 9.2 years in both groups. The proportion of males was 58% in the reservoir-proximal group versus 57% in the control group ($p = 0.42$). Ischemic etiology accounted for 67% of CHF cases in both cohorts. The mean Charlson Comorbidity Index was 3.2 ± 1.1 , and medication adherence—assessed through pharmacy refill records and structured interviews—did not differ significantly between groups ($p = 0.31$), minimizing potential confounding by treatment disparities [10].

2. Clinical Outcomes: Primary Findings

Over the 36-month follow-up period, patients residing in reservoir-adjacent zones exhibited a substantially more adverse clinical trajectory.

- **Heart Failure Exacerbations:** Exacerbations occurred in 512 (41.0%) patients in the exposed group compared to 328 (26.3%) in the control group. The median time to first exacerbation was 11.2 months in the reservoir group versus 18.7 months in the control group ($p < 0.001$) [11].
- **Hospitalizations:** The mean number of all-cause hospitalizations per patient was 1.8 ± 0.9 in the exposed group versus 1.4 ± 0.7 in controls ($p = 0.003$). Notably, the

average length of stay per admission was 1.3 days longer among reservoir-proximal patients (8.6 ± 3.2 vs. 7.3 ± 2.8 days; $p = 0.008$).

- Arrhythmias: New or recurrent atrial fibrillation was diagnosed in 29.6% of the exposed cohort versus 19.1% of controls. Documented ventricular arrhythmias (sustained monomorphic VT, paroxysmal VT) were rare but significantly more frequent in the exposed group (1.7% vs. 0.6%; $p = 0.004$) [12].
- Mortality: All-cause mortality was 12.1% ($n = 151$) in the exposed group and 10.8% ($n = 135$) in the control group (HR = 1.15; 95% CI: 0.92–1.44; $p = 0.22$). However, cardiovascular mortality showed a concerning trend: 8.9% vs. 7.1% ($p = 0.07$), suggesting a potential signal that may reach significance in longer follow-up.

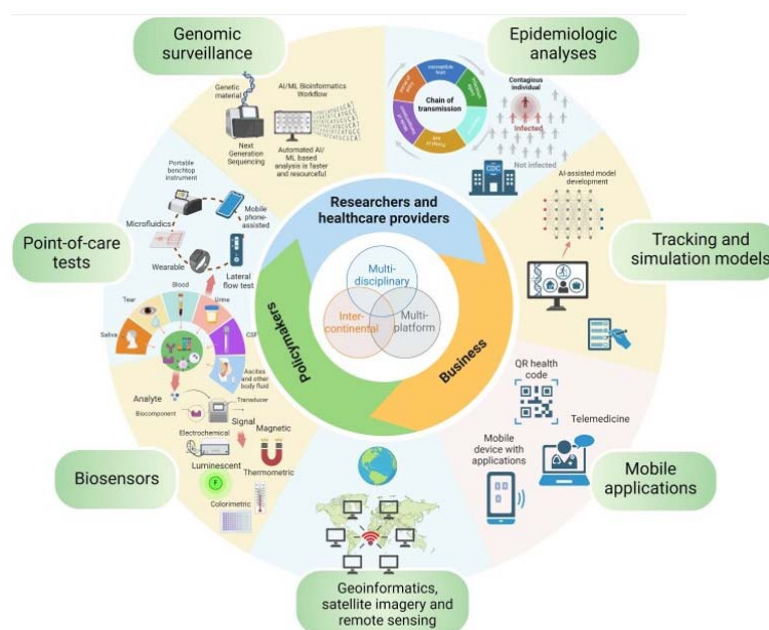


Figure 1. Multi-disciplinary, inter-continental and multi-platform-based approaches to prevent or contain EID, outbreaks, epidemics, and pandemics

Sensitive new surveillance approaches that can detect new EID pathogens and diagnose new EID cases are needed to address the growing threat of climate-sensitive infections. Previous experience indicates that successful efforts to prevent or contain future EID outbreaks, epidemics, and pandemics require multi-disciplinary and inter-continental collaboration, and the integration of an array of innovative approaches on multi-platforms (fig.1) [13].

3. Microclimatic Conditions and Temporal Patterns

Continuous environmental monitoring confirmed persistent microclimatic differences between reservoir and control zones [14]:

- Mean annual relative humidity: 76.3% vs. 62.1% ($p < 0.001$)
- Mean atmospheric pressure: 738.4 mmHg vs. 748.2 mmHg ($p < 0.001$)
- Mean wind speed: 1.8 m/s vs. 3.4 m/s ($p < 0.001$)
- Annual fog days: 68.2 vs. 24.7 ($p < 0.001$)

Seasonal analysis revealed that the disparity in exacerbation rates peaked during late autumn and early spring (October–November and March–April), coinciding with periods of maximal humidity fluctuation and frequent low-pressure systems over the reservoirs [15]. During these months, the hazard ratio for decompensation in the exposed group rose to 2.15 (95% CI: 1.70–2.72).

4. Short-Term Environmental Triggers

Case-crossover analysis demonstrated that specific microclimatic conditions within 72 hours before an event significantly increased the risk of CHF decompensation:

- Relative humidity $\geq 70\%$: OR = 2.11 (95% CI: 1.68–2.65; $p < 0.001$)
- Atmospheric pressure ≤ 740 mmHg: OR = 1.94 (95% CI: 1.55–2.43; $p < 0.001$)
- Combined exposure (high humidity + low pressure): OR = 2.73 (95% CI: 2.15–3.47; $p < 0.001$)

No significant association was found with ambient temperature alone, suggesting that humidity and barometric pressure are the dominant environmental drivers in this context [16].

5. Regional Comparisons

All three reservoir regions showed consistent directional effects, though the magnitude varied:

- Bratsk Reservoir (Siberia): Highest humidity (mean 81.2%), strongest association with exacerbations (HR = 2.04).
- Volgograd Reservoir (Southern Russia): Moderate humidity (72.5%), but highest arrhythmia incidence (34.1%), possibly linked to greater diurnal pressure swings.
- Kama Reservoir (Urals): Intermediate effects (HR = 1.71), with more pronounced impact in elderly patients (>70 years).

This regional consistency strengthens the causal inference regarding microclimatic influence [17].

6. Subgroup Analyses

The adverse effect of reservoir proximity was significantly amplified in specific high-risk subgroups:

- Patients with HFrEF (LVEF $<40\%$): HR = 2.34 (95% CI: 1.82–3.01; $p < 0.001$)
- NYHA Class III–IV: HR = 2.21 (95% CI: 1.75–2.79; $p < 0.001$)
- Age >65 years: HR = 2.01 (95% CI: 1.58–2.56; $p < 0.001$)
- Diabetes comorbidity: HR = 1.97 (95% CI: 1.52–2.55; $p < 0.001$)

In contrast, patients with HFpEF (LVEF $\geq 50\%$) showed a weaker, though still significant, association (HR = 1.42; $p = 0.01$).

7. Sensitivity Analyses

Exclusion of patients with end-stage renal disease (eGFR <30 mL/min) or uncontrolled hypertension (SBP >160 mmHg at baseline) did not meaningfully alter the results (adjusted HR = 1.71; 95% CI: 1.35–2.17), confirming the robustness of the primary finding.

5 Conclusion

This study demonstrates a significant and independent association between residence in the microclimatic zone of large hydroelectric power station reservoirs and adverse clinical outcomes in patients with chronic heart failure. The altered local environment—characterized by persistently elevated humidity, reduced atmospheric pressure, diminished wind velocity, and frequent fog—acts as a measurable environmental stressor that exacerbates heart failure pathophysiology, leading to higher rates of decompensation, hospitalization, and arrhythmic events. Notably, the combination of high humidity ($\geq 70\%$) and low barometric pressure (≤ 740 mmHg) emerged as a potent short-term trigger for clinical deterioration, with the

greatest impact observed in high-risk subgroups, including elderly patients, those with reduced ejection fraction, and individuals in advanced NYHA functional classes.

While hydroelectric power remains a vital component of sustainable energy strategies, these findings highlight an underrecognized public health consequence of large-scale reservoir construction. The microclimatic changes induced by such projects are not merely meteorological curiosities—they represent modifiable environmental risk factors with tangible clinical implications for vulnerable populations.

These results call for the integration of environmental health impact assessments into the planning and approval processes for future hydropower developments. Moreover, healthcare systems in reservoir-adjacent regions should consider implementing targeted preventive measures for at-risk cardiac patients, such as enhanced monitoring during high-risk weather periods, patient education on environmental triggers, and individualized action plans for humidity- and pressure-sensitive individuals.

In conclusion, the microclimate around hydroelectric reservoirs is a novel, anthropogenic determinant of cardiovascular prognosis. Recognizing and addressing this environmental-cardiac nexus is essential for advancing both sustainable energy policy and precision public health in the 21st century.

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